| Qn | Ans | Solution |
|---|---|---|
| 19 | C | $E_q = \frac{q}{4\pi\varepsilon_0 r_q^2} = \frac{q}{4\pi\varepsilon_0 (0.40)^2} = 12 \text{ N C}^{-1}$ → $\frac{q}{4\pi\varepsilon_0}$ = 12(0.40)$^2$ = 1.92<br>$E_{\text{total}} = E_q + E_{-q}$<br>$= \frac{q}{4\pi\varepsilon_0 (0.40)^2} + \frac{q}{4\pi\varepsilon_0 (0.60)^2}$<br>$= 12 + \frac{1.92}{0.60^2}$<br>$= 17 \text{ N C}^{-1}$ |
| 20 | A | Temperature of thermistor increases with increasing current $I$ |